Where do libraries come from, and where do they stop helping?

Bootstrapped libraries, documentation, evidence selection, and boundaries

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In one sentence

Small models handle inputs they cannot read when code holds the work: a 0.6B caller runs an 8B-written MRCR library to 1M tokens; a trained 4B with three labels met a pre-registered 10-point non-inferiority margin against an 8B with thirty (75 vs 73 of 141, not under template clustering); bootstrapped libraries break when the question is reworded, and a 4B question parser repairs them. Recursion is not the active ingredient; acquisition is the open problem.

Libraries can be bootstrapped: an 8B wrote MRCR's library.

MRCR held out (8k to 1M tokens)	2 needles	4 needles
Direct answer, 4B	0.23	0.15
RLM, untrained 4B, no library	0.18	0.11
8B-written library, run alone	1.000	1.000
0.6B caller + library	**1.000**	**1.000**
Untrained 4B + library	0.95	0.88

64 rows per column. The 0.6B caller never saw MRCR. The proposer saw the contract and three solved examples; candidates run on 30 teaching tasks; the best scored 30/30 and was reviewed before use (40 lines of parsing, nothing copied from the examples).

Who can write it (seeds accepted)

4B Instruct: 0 of 1
4B, thinking off / on: 0 of 1 / 0 of 2
8B, thinking off: 0 of 2
8B, thinking on: **2 of 5**
14B (4-bit), thinking on: **2 of 5**
4B after SFT + RL: 0 of 4; test-time RL on MRCR: 0 in 1,920 samples

Caveat: MRCR is solvable by a short parser, so this measures program discovery, not reading. We found no published note saying so.

Ledger E21, E21b; runs e21-*. MRCR: openai/mrcr (MIT), official SequenceMatcher grader.

Discovery on 37 generated families: thinking, scale, and training.

Proposer	Solved / 37	Tokens
4B Instruct (2 seeds)	5, 4	4k
8B, no thinking	4	4k
4B, thinking	14	55k
8B, thinking (2 seeds)	20, 25	48k
14B, thinking (2 seeds)	21, 26	30k
4B + SFT (2 seeds)	9, 7	3k
4B + SFT + RL (2 seeds)	16, 14	2k
4B, round four (2 seeds)	**22, 20**	2k
avg. of 4 RL seeds	15.8	
4B, 3x helper data, avg.	16.8	2k

Tokens: proposer output per family. Families vary record format, question and answer format; held-out families use a new format or question.

Training the proposer: SFT on the 8B's verified code, then 40 steps of RL with the verified teaching score as reward: 9 to 30 of 74 attempts ($p < 0.001$). Seed-sensitive: round four's 42 (8B 45) is the best of four RL seeds (22 to 42). MRCR still fails.

Helper gap: 3x the helper teacher data matches round four on solves, reaches the 8B on helper families (mean 0.46), and varies far less by seed.

Ledger E26 to E26g, E36 to E46; runs e26-disc-*, e4*. Generator: family_gen.py (answers exact).

Bootstrapping is cheap where answers are exact (pre-registered).

	Correct	False acc.	Tokens
E39: 30 new families, 2 seeds			
4B, best of 8, 3 labels	12, 14	0, 1	3.7k
8B thinking, 30 labels	13, 16	0, 0	52.4k
E58: 141 untouched families (powered)			
4B, best of 8, 3 labels	**75**	10	21k
8B thinking, 30 labels	73	0	64k

Correct = 0.9+ on 30 fresh tasks; tokens per family (E39: output only). E58 met the registered 10-point non-inferiority margin with families as the unit (95% CI -0.078 to $+0.106$), not with templates as the unit; the weakest checkpoint did not (67 vs 73).

Why three labels suffice

Candidates that run cleanly and reproduce the prompt's examples are correct: precision 0.81, 0.88, 0.98 with 1, 2, 3 labels (127 of 129 at three, all generalizing). Agreement among candidates is no substitute: an RL-trained proposer repeats its wrong programs.

Limits: noisy families need tens of labels; E58's false acceptances: all 6 semantic totals, 3 of 7 day spans.

Ledger E37 to E39, E58; notebook assets-e39- and assets-e58-preregistration.md.

Libraries generalize in length, not in wording; the harness can fix that.

Unseen shift (of 55)	As is	4B	0.6B
Inputs 5× longer	55		
Question reworded	3	**54**	22
Prefix, field order	11	**48**	26

The parser wrapper: a 4B reads only the question, returns the family's parameters (96% exact on unseen rewordings), and the question is re-rendered in the original template; the frozen library runs unchanged. Remaining failures are structured formats whose records changed.

Who can supply the robust interface?

4B proposer, told how, writing from scratch: 1 of 11 and 0 of 18 accepted libraries adopt it.
4B editing a working library: working edits for 3 of 12 families (2 robust).
Hand-written wrapper + 4B parser: 54 of 55.

Executors generalize; a 4B parses well; a small proposer rarely writes the interface. Acquisition is the open problem.

Ledger E59 to E61, E68, E69; stress_gen.py, e68_canon.py.

One line of documentation decides whether an untrained caller uses a library.

MRCR, untrained 4B caller, docstring	Score
Signature only	0.18
Description + signature (as the 8B wrote it)	0.59
Signature + usage line	0.95
Description + signature + usage line	0.95
Hand-written library, usage line removed	0.22
Hand-written library, as written	0.95

Usage line: `Typical use: FINAL_TEXT(answer_mrcr(request)).`

Without it the untrained 4B hands the whole input to a sub-model (overflowing at 128k and above) or retypes the routine's output by hand.

The distilled 0.6B scores 1.000 without the line; the untrained 0.6B fails either way (0.21, 0.13). Robustness comes from distillation, not size. The bootstrapper now inserts the line.

Ledger E25c, E25d. One family, one library, one seed per arm.

MuSiQue in long haystacks: capacity belongs in choosing evidence.

Answer F1 (90 to 180 held-out questions per row)	8k	32k	128k	tokens
Direct answer, 4B	0.52	0.25	overflow	
RLM without a library, depth 0 / depth 1	0.49 / 0.51	0.25 / 0.25	overflow	
Library that hands each hop to a child RLM	0.29	0.29	0.34	3 to 4 times
0.6B caller + filter, BM25, read	0.52	0.31	0.46	57k
0.6B caller + two-round retrieval	**0.53**	**0.39**	**0.50**	10k

What helps: a second, targeted retrieval (BM25 for the question, the helper names the missing fact, BM25 again, one read): +0.058 over BM25 alone, +0.045 over filter+BM25 at a sixth of the tokens.

What does not: recursion; an 8B reader (+0.01; gold paragraphs reach 0.63 to 0.67 with either reader); more sampling (majority of 3 at 88k tokens: 0.22 vs 0.35 for the library at 32k).

Ledger E22c to E22f, E23, E25e, E28. MuSiQue (CC BY 4.0) dev questions in haystacks.

External validity, and what we withdrew.

LongBench-v2 (all 503 questions, chance 0.25)

Direct (truncated): 0.356, 42k tokens
Retrieve-then-read: 0.427, 10k tokens
via the 0.6B caller: 0.423 (+0.07, se 0.02)

Selecting evidence beats summarizing (map-reduce: no gain).

Boundary: no proposer bootstraps OOLONG, our own family (8B 0.20, 14B 0.00; shipped library 0.60): 30 examples span 11 question templates.

Withdrawn: distractor-limited reader; length routing (pre-registered); "no help on LongBench-v2" (it was map-reduce).

Method lessons

- Verify under the semantics you run under: the kernel's `ask` raises on a cut-off reply, scripted harnesses returned partial text (one routine crashed in 46 of 90 caller runs).
- Verification must span deployment lengths: 8k-only teaching selected a pass-through routine that scores 0 at 128k.
- Routine evolution (GEPA-style): prompt rewrites by the 8B hurt; numbers helped (+0.048, ns); random search found nothing.

Ledger E32 to E32e, E31, E22b, E24, E24c, E35. LongBench-v2: zai-org (Apache-2.0).

What this changes, and what to test next.

The thesis, revised. For small models on large inputs, capability comes from code with programmatic access and tested routines, plus narrow neural decisions (parse a question, classify an item). Recursion is not the active ingredient; we keep the term RLM and frame it as a harness. The mechanisms are prior art (CodeAct, RLM, skill libraries, FlashExtract, Evaporate); what is new is measured on small models: acquiring robust routines cheaply, and where that breaks (wording, lengths beyond teaching, false acceptance).

Next

1. The deciding experiment: interface (tool loop vs REPL) $\times$ source of procedures (primitives, human-written, acquired), plus a host pipeline with no controller. Is the thesis about the harness or acquisition?
2. False acceptance without labels: metamorphic checks (shift times, keep dates) and helper consistency.
3. Who writes the interface: an 8B proposer (running), or the harness supplies it.
4. Acceptance must span deployment lengths and wordings (E52b, E57, E59).

Appendix: protocol and provenance.

Models: Qwen3-4B-Instruct-2507 (helper, caller), Qwen3-0.6B + LoRA (distilled caller), Qwen3-4B hybrid and Qwen3-8B-AWQ (proposers), all served with vLLM 0.27.1 on two A100 MIG 7g.40gb slices.

Data: OpenAI MRCR (MIT; within-bin split, 30 teaching rows, 64 held out per needle count); MuSiQue v1.0 (CC BY 4.0; teaching from train, held out from dev; two samples of 90); LongBench-v2 (Apache-2.0, revision 2b48e494); generated families (family_gen.py, exact answers).

Statistics: paired differences with standard errors over all questions; exact McNemar for family counts; the routing test was pre-registered.

Every number above has a ledger row in docs/RESULTS.md and a run directory in the run store; the lab notebook (docs/notebook/2026-10-19-...) records failures and corrections in time order.